

Rahul Rawat

BUSINESS ANALYST WERKSTUDENT

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PROFILE

Data Science and Analytics Master student at SRH Heidelberg based in Mannheim, with hands on business analysis and process automation work across LLM driven agents, cloud pipelines, and BI dashboards. I have shipped a Retrieval Augmented Generation system with agentic routing over Llama 3.1 8b via Groq, a Random Forest driven study translating raw event data into business relevant risk insights, and a fully automated BigQuery medallion pipeline with a five page Looker Studio dashboard. Comfortable in Python, SQL, and workflow thinking, I am the right fit for analysing business processes, identifying automation potential, and supporting the technical rollout of workflow automations.

EXPERIENCE

eRay GmbH

Oct 2025 to Mar 2026

Data Scientist

- Built an end to end recursive time series pipeline forecasting chlorophyll a, turbidity, pH, and dissolved oxygen for a German lake, delivered as a six month collaboration with SRH University Heidelberg.
- Benchmarked six models head to head, Ridge, Gradient Boosting, LightGBM, XGBoost, CatBoost, and Prophet, and used CatBoost multi quantile regression to produce asymmetric 80 percent prediction intervals for decision support under uncertainty.
- Enforced strict anti leakage rules across the pipeline, surfacing the honest finding that pH and dissolved oxygen are physically predictable while chlorophyll a and turbidity are not without live optical sensors.
- Reconstructed missing winter readings with MICE imputation and engineered a synthetic winter decay forecast canvas so tree based models stopped flatlining during recursive prediction.
- Wrapped the whole pipeline in an orchestrator with gate checks and velocity and ecological bounds that halts on failed imputation rather than letting bad data cascade downstream.

EDUCATION

M.Sc. Data Science and Analytics

Apr 2025 to Present

SRH University of Applied Sciences Heidelberg

Bachelor of Technology in Computer Science

2019 to 2023

GL Bajaj Institute of Technology and Management

PERSONAL PROJECTS

Hybrid RAG Orchestrator with Agentic Routing

Python

LangChain

Llama 3.1 8b via Groq

ChromaDB

HuggingFace MiniLM L6 v2

Streamlit

- Delivered a modular Retrieval Augmented Generation system that classifies user intent into three execution paths, local knowledge retrieval, external web search, or direct conversational logic, measured by clean end to end responses across all three routes, using a custom decision making router on top of Llama 3.1 8b via Groq and LangChain orchestration.
- Built a persistent semantic memory over PDF and vector data, measured by consistent retrieval on cross session queries, using ChromaDB with local persistence and HuggingFace MiniLM L6 v2 embeddings.
- Kept multi turn context intact for the LLM, measured by coherent follow up answers rather than isolated single shot responses, by adding a stateful MemoryAgent directly into the inference pipeline.
- Shipped the whole system as a Streamlit interface with end to end ownership, measured by a working deployed prototype, using Groq inference and a modular routing layer that can be extended per new intent.

Economic Impact Analysis of Global Climate Events

Python with Pandas and scikit learn

Random Forest

statistical modelling

Matplotlib and Seaborn

- Executed an end to end data science project transforming raw global event datasets into structured insights for resource allocation and risk assessment, measured by a single reproducible pipeline from raw CSV to a stakeholder report.
- Performed advanced data preparation and cleansing over outliers, imputation, and normalisation, measured by stable model performance before and after feature scaling, to build a high quality data foundation.
- Developed Random Forest models to analyse correlations between event duration and financial impact, measured by feature importance rankings and residual analysis, extracting clear business relevant insights on where economic risk concentrates.
- Communicated complex statistical findings to non technical stakeholders, measured by comprehensive visual reports and calibrated confidence statements that could survive a management review.

Movie Analytics and ML Pipeline on GCP

GCP BigQuery

Cloud Run

GCS

Cloud Scheduler

BigQuery ML

Python

SQL

Looker Studio

- Built an end to end batch pipeline pulling movie data from a public API into a GCS data lake and processing it through Bronze to Silver to Gold medallion architecture in BigQuery, measured by a fully automated Cloud Scheduler trigger with no manual intervention required, using Cloud Run for compute.
- Engineered the Silver layer for trustworthy analytics with schema enforcement, safe type casting, deduplication via window functions, and genre normalisation into a relational model, measured by clean referential integrity across all downstream Gold tables.
- Trained a BigQuery ML classifier that predicts whether a film will be a hit before release, measured by leakage free evaluation, by deliberately splitting features into two tables so only pre release signals feed the model.
- Built Gold layer aggregates and a five page Looker Studio dashboard answering concrete business questions on genre ROI, foreign language growth, and release season timing, measured against the ML early warning view, and secured the system with a least privilege service account and Secret Manager.

RESEARCH AND THESIS

Bachelor Thesis, Diabetes Prediction Using Machine Learning

GL Bajaj Institute of Technology and Management, IEEE style paper

- Built a full end to end machine learning pipeline comparing six classifiers on a clinical dataset of 768 patients, using 10 fold cross validation and per model confusion matrices to make the model comparison auditable.
- Caught biologically impossible zero values that the original authors had overlooked, then applied IQR based outlier removal and proper imputation to restore data integrity before any model fit.
- Chose ROC AUC over accuracy as the headline metric for a 65 to 35 imbalanced dataset, avoiding a misleadingly rosy accuracy figure that would have hidden real error patterns in the minority class.
- Wrote up findings as an IEEE style paper including an honest section on limitations and what to do differently in a follow up study, publishable in substance.

CERTIFICATIONS

- Google Data Analytics: Foundations, Data, Data, Everywhere, issued April 2025 on Coursera.
- SAS Certified Specialist: Visual Business Analytics Using SAS Viya, issued May 2025.
- NVIDIA: Building LLM Applications With Prompt Engineering, issued November 2025.

ACHIEVEMENTS

- USAII Global AI Hackathon 2026: Finalist at Graduate Level, awarded by the United States Artificial Intelligence Institute for innovation, technical creativity, and applied AI on real world challenges.

LANGUAGES

English: fluent, professional working proficiency. **German:** B1, in progress.